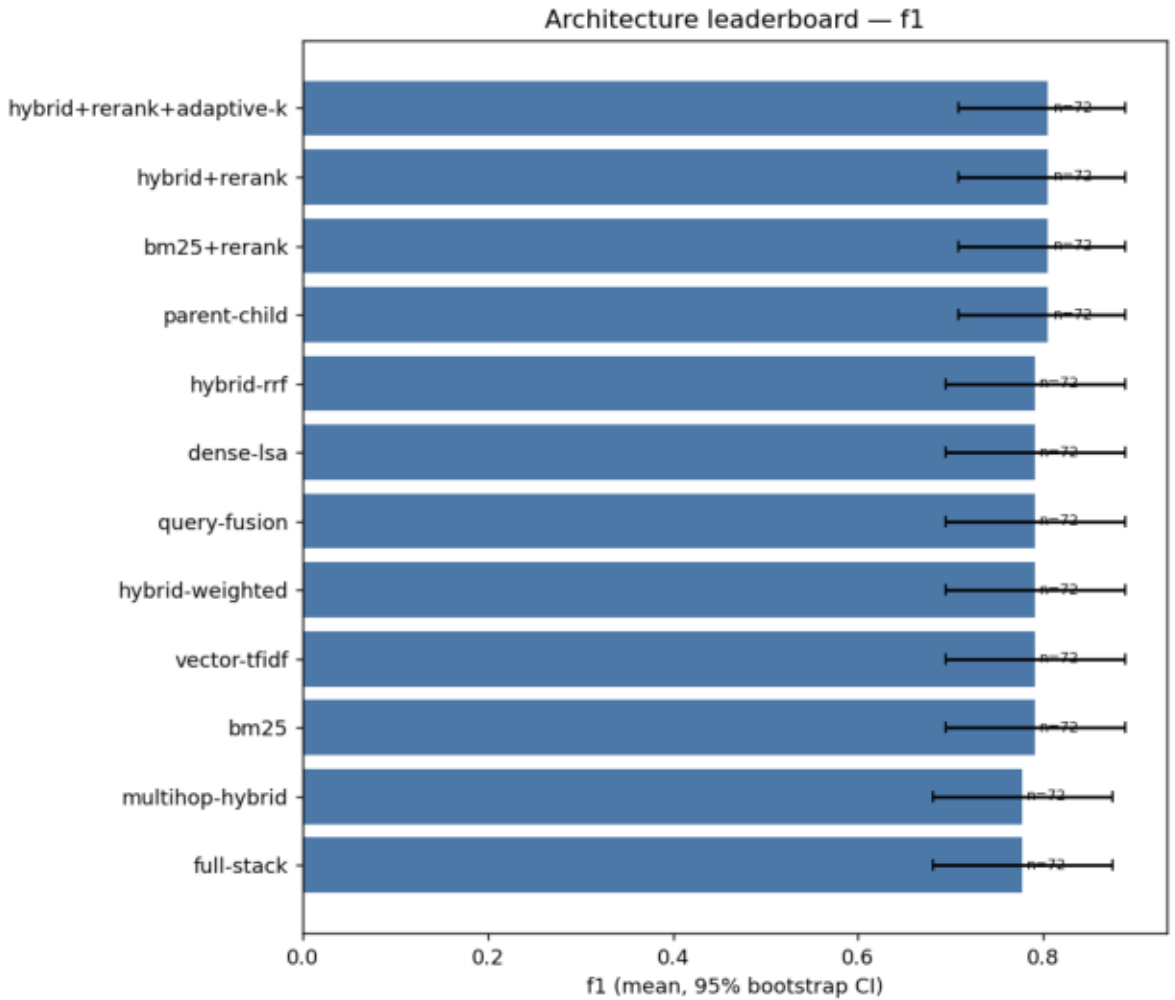
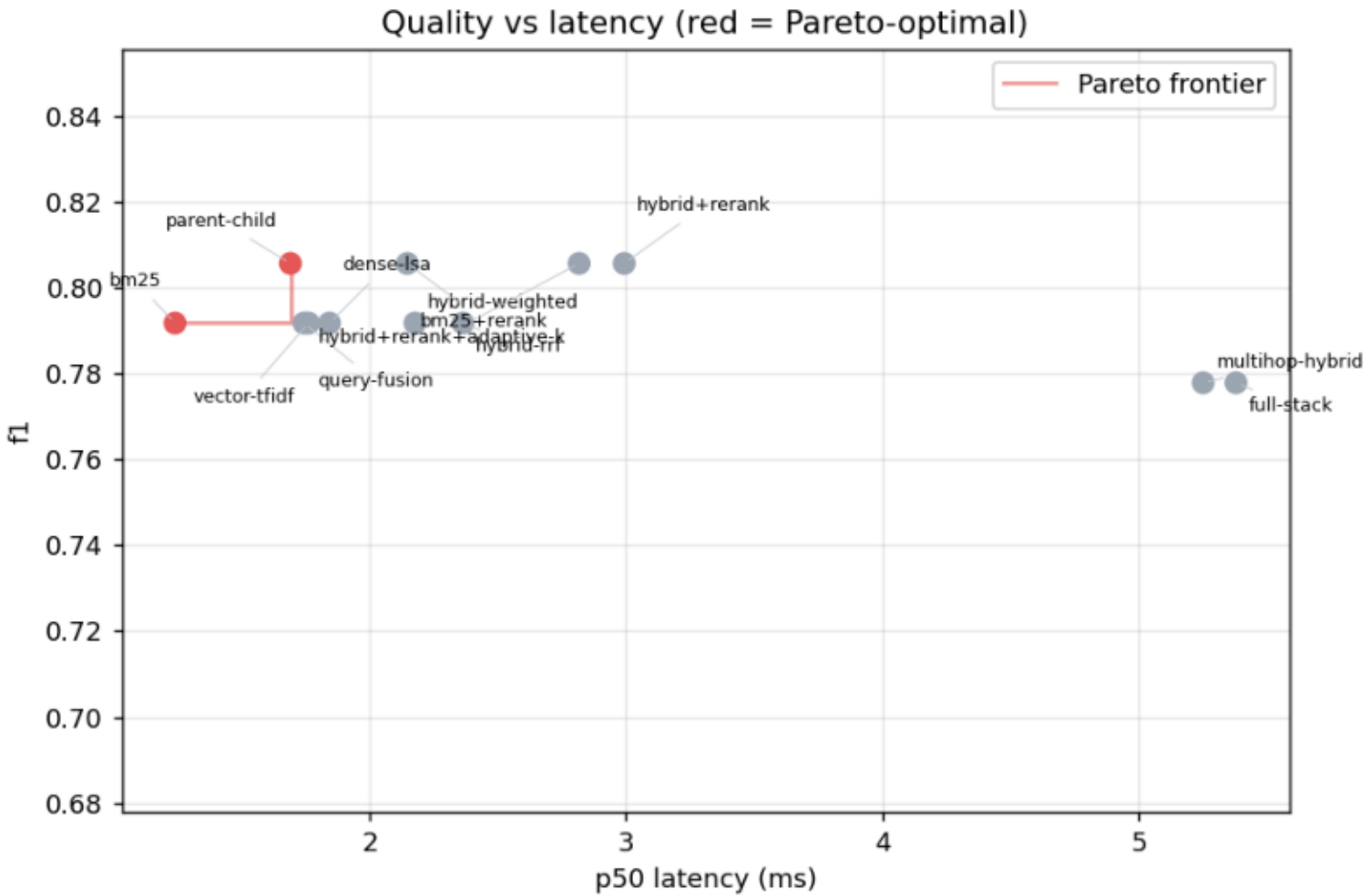
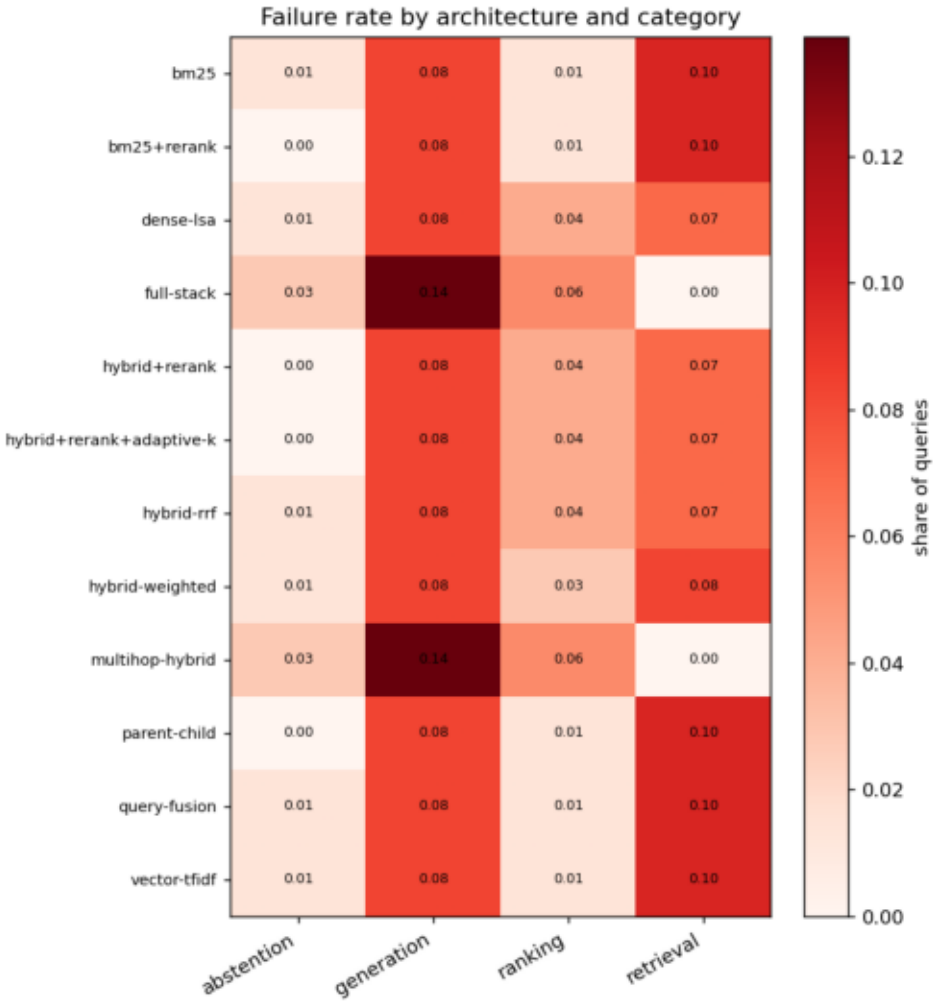
1. Architecture leaderboard (held-out test split)

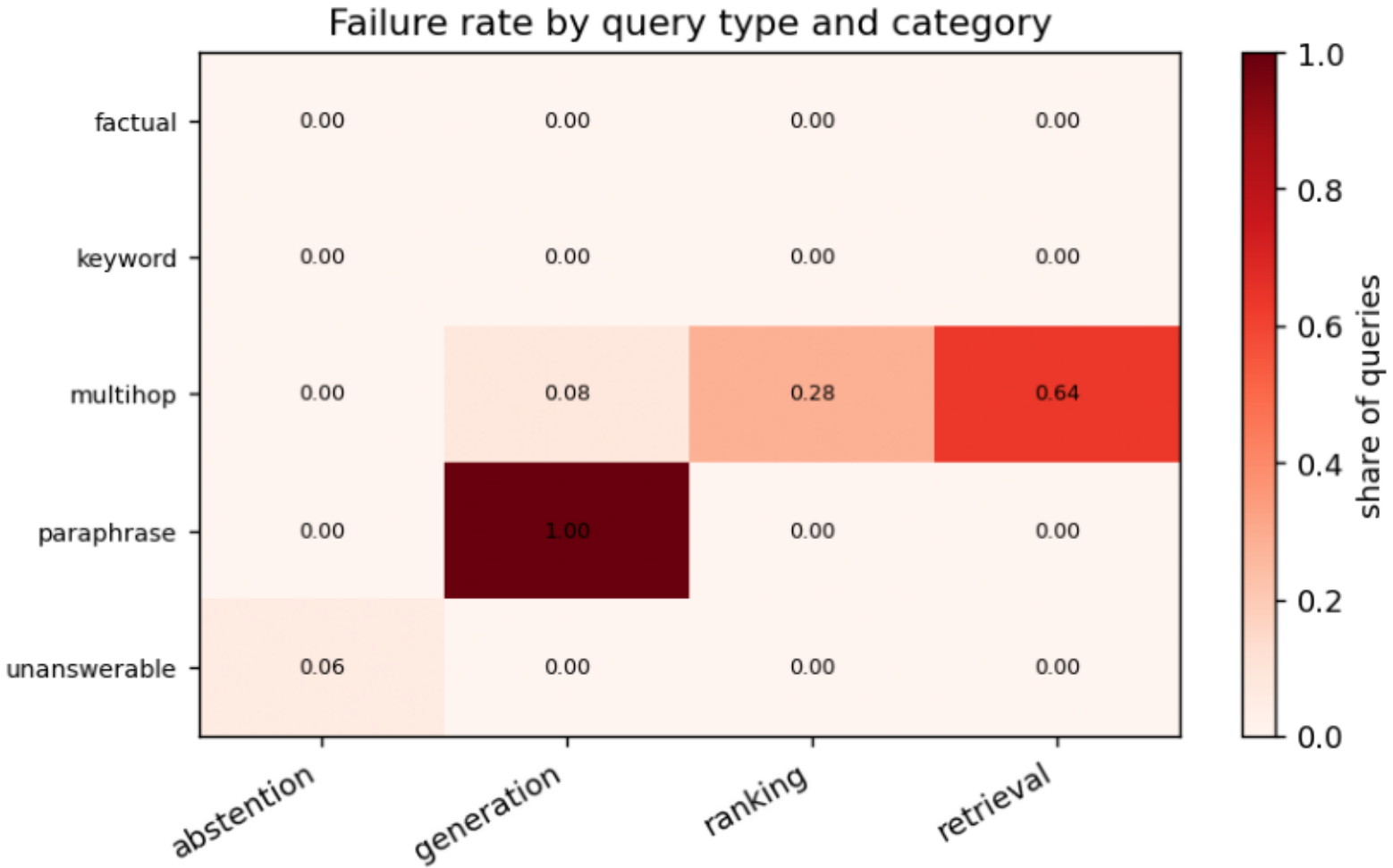


2. Quality vs latency (Pareto frontier)

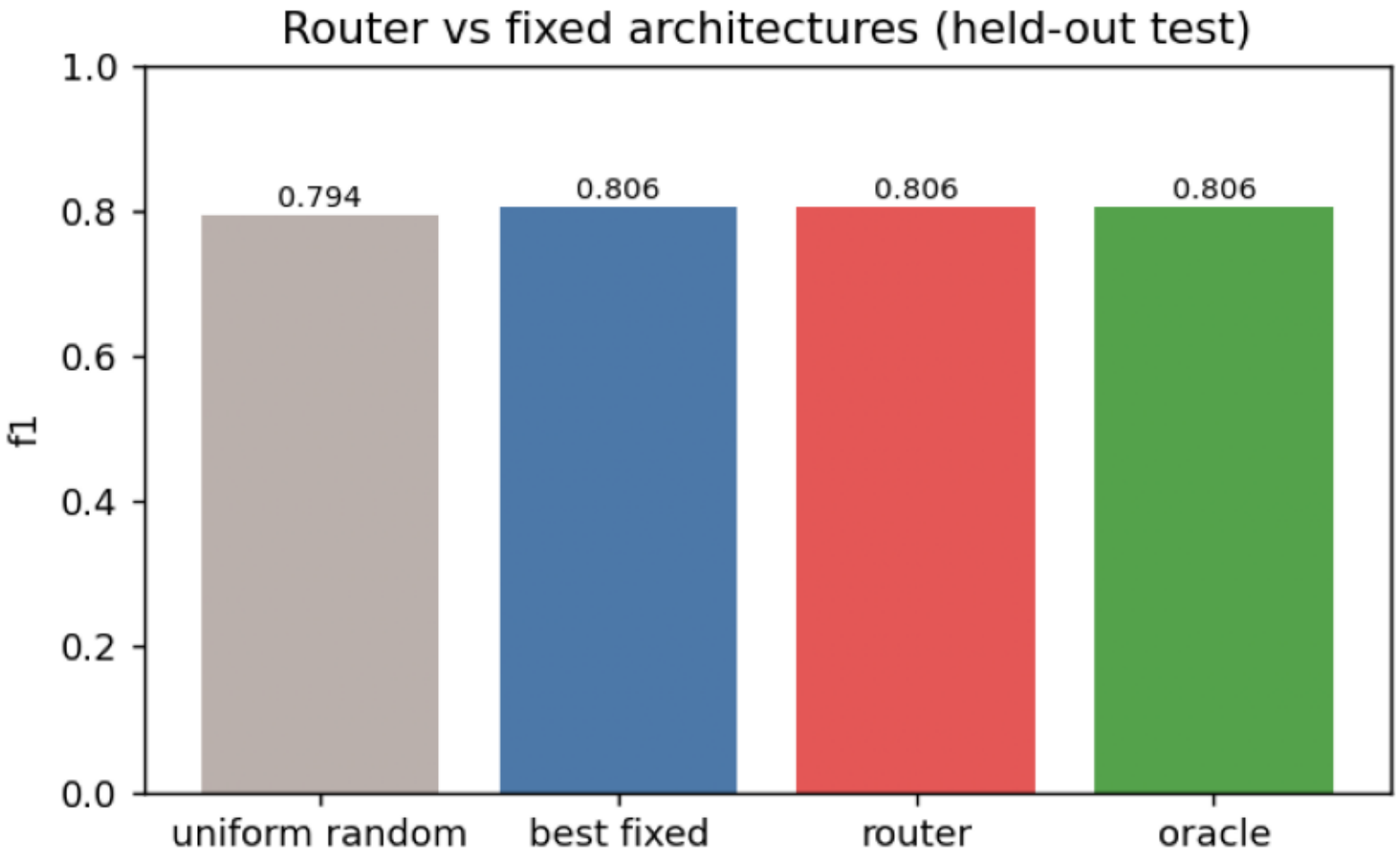


5. Failure diagnosis

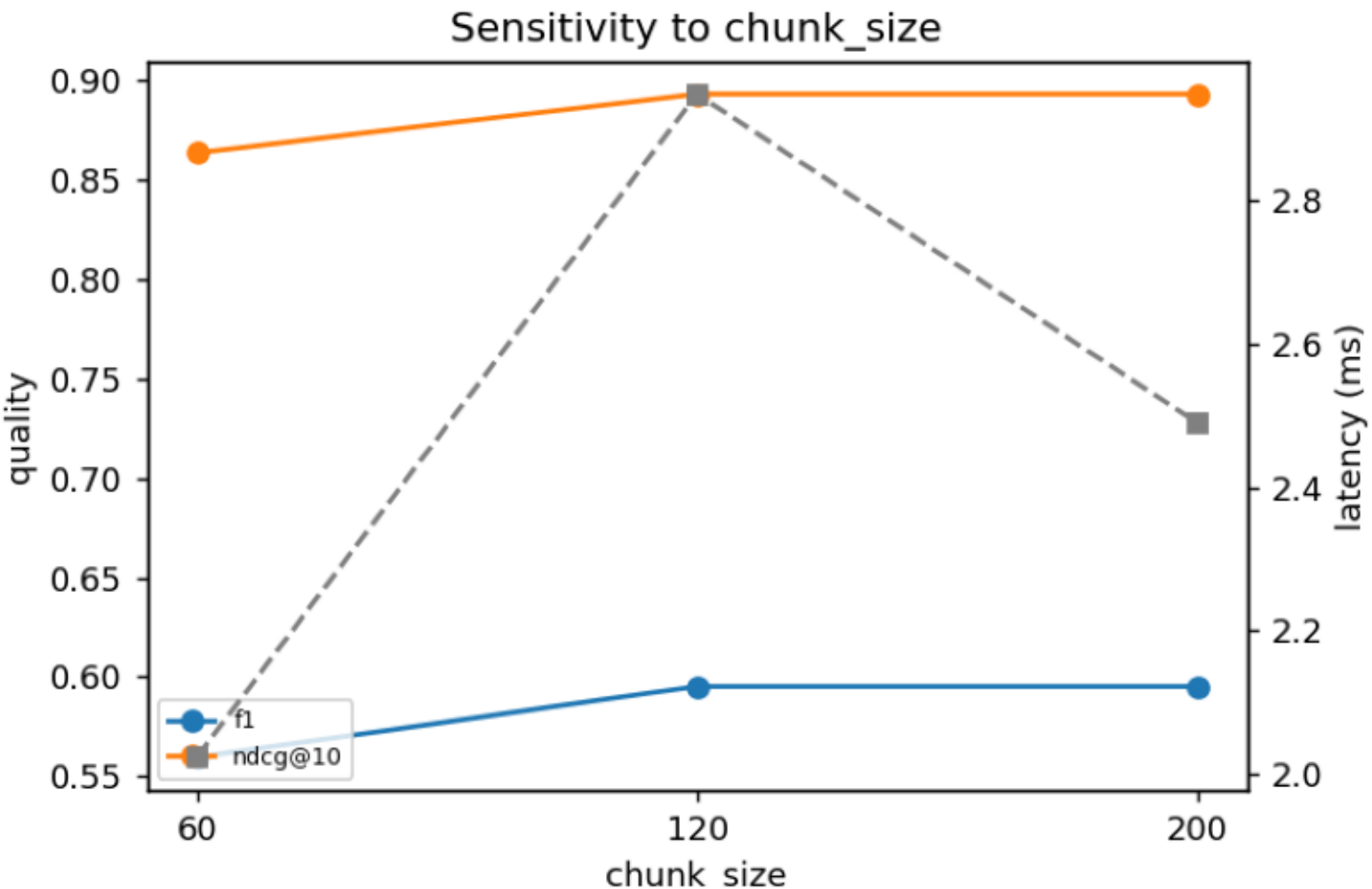



Failure rate by query type and category
factual
keyword
multihop
paraphrase
unanswerable
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.08
0.28
0.64
0.00
1.00
0.00
0.00
0.06
0.00
0.00
0.00
abstention
generation
ranking
retrieval
share of queries
1.0
0.8
0.6
0.4
0.2
0.0

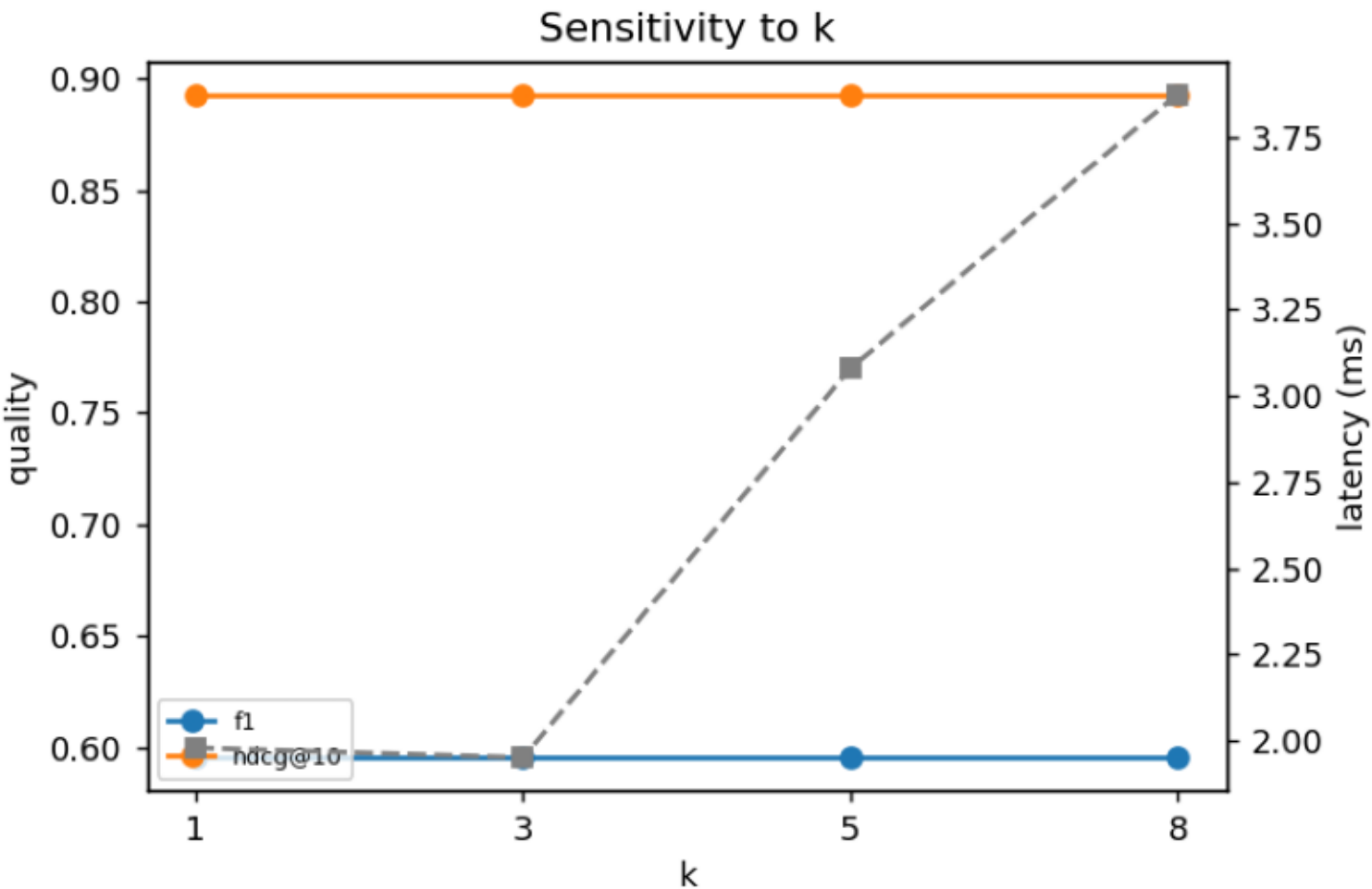
6. Query-aware router



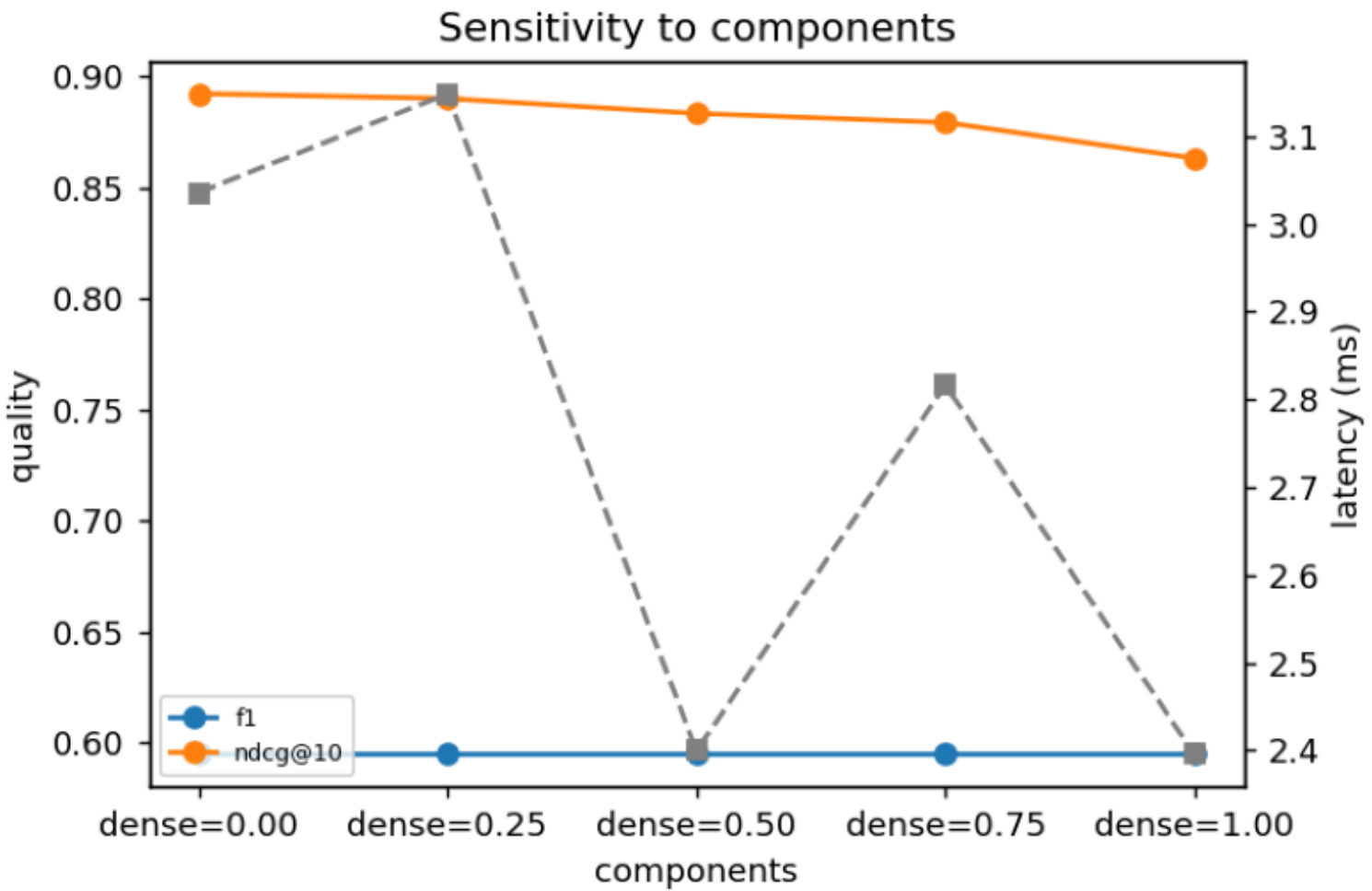
8. Sweep: chunk_size



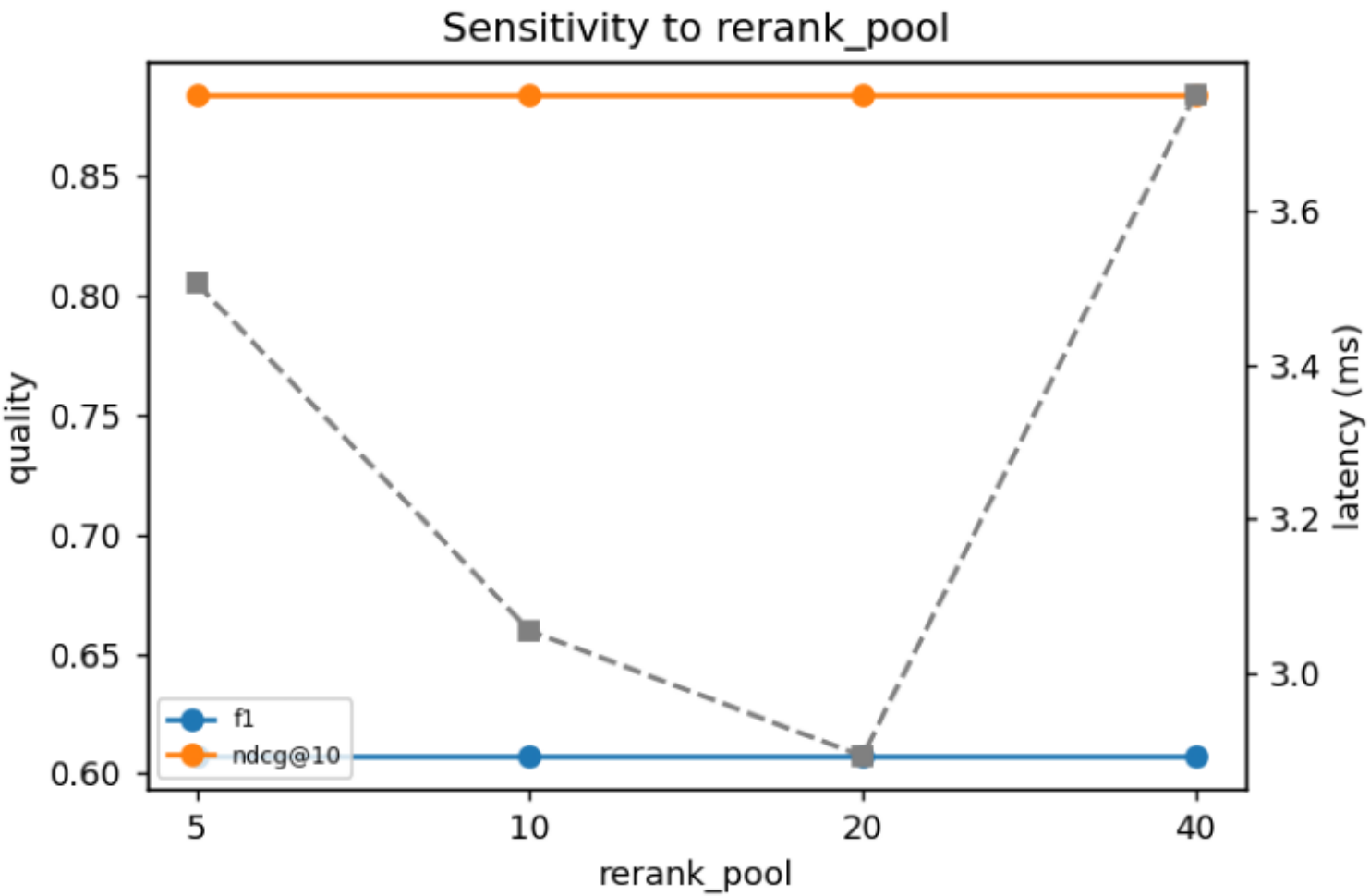
8. Sweep: top_k


Sensitivity to k
quality
latency (ms)
k
0.90
0.85
0.80
0.75
0.70
0.65
0.60
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00
1
3
5
8
f1
ndcg@10

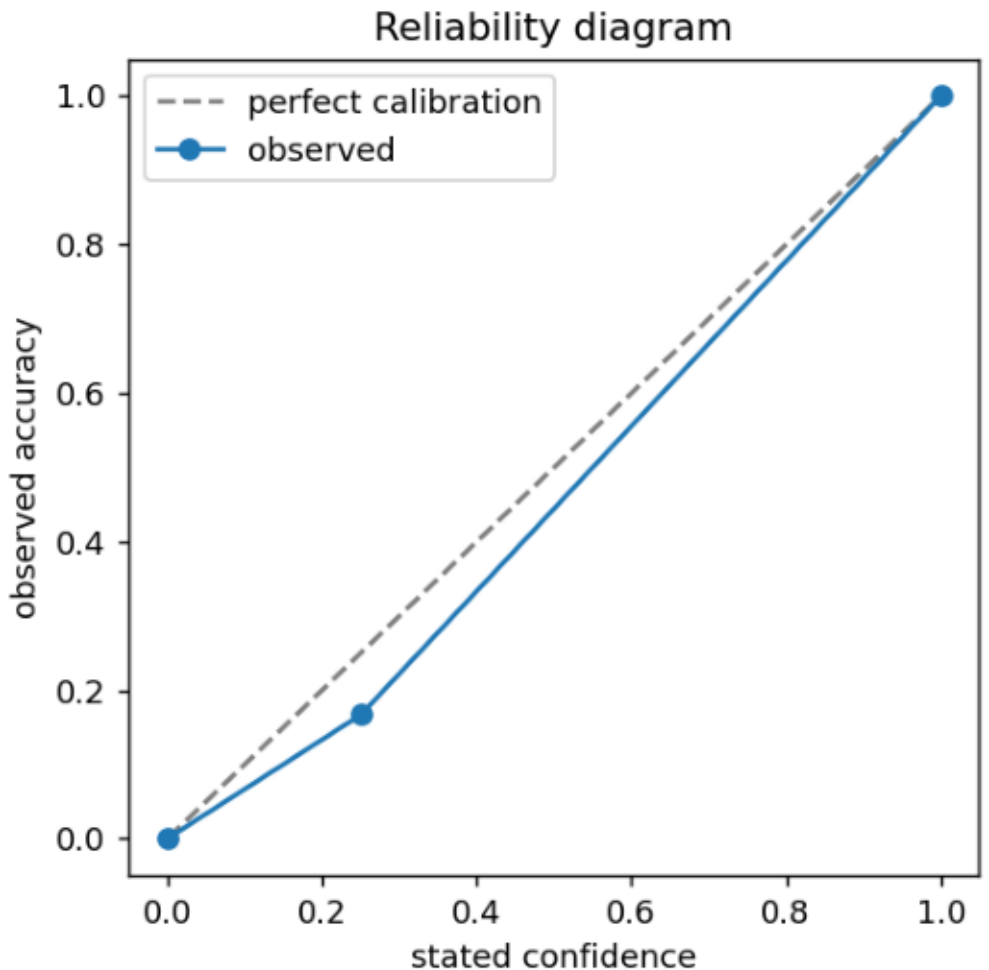
8. Sweep: hybrid_weight



8. Sweep: rerank_pool



9. Confidence calibration



10. Caching: none vs exact vs semantic

